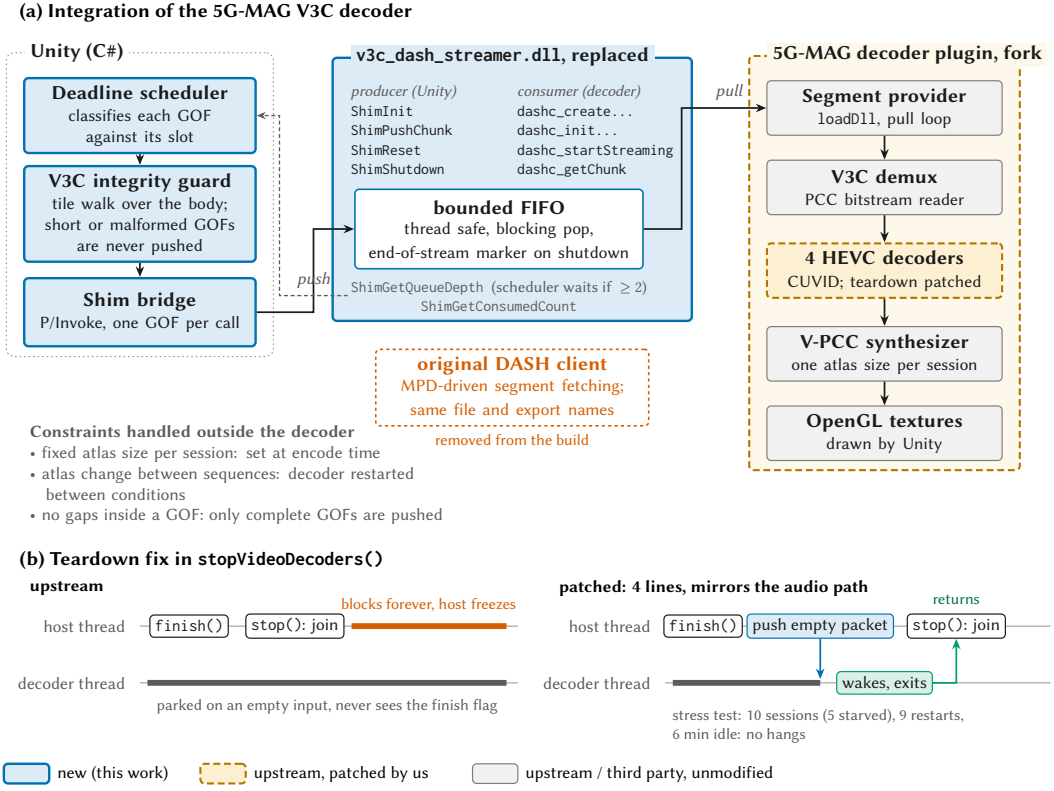
(a) Integration of the 5G-MAG V3C decoder
Unity (C#)
Deadline scheduler
classifies each GOF against its slot
V3C integrity guard
tile walk over the body; short or malformed GOFs are never pushed
Shim bridge
P/Invoke, one GOF per call
push
v3c_dash_streamer.dll, replaced
producer (Unity)
ShimInit
ShimPushChunk
ShimReset
ShimShutdown
consumer (decoder)
dashc_create...
dashc_init...
dashc_startStreaming
dashc_getChunk
bounded FIFO
thread safe, blocking pop, end-of-stream marker on shutdown
ShimGetQueueDepth (scheduler waits if ≥ 2)
ShimGetConsumedCount
pull
5G-MAG decoder plugin, fork
Segment provider
loadDll, pull loop
V3C demux
PCC bitstream reader
4 HEVC decoders
CUVID; teardown patched
V-PCC synthesizer
one atlas size per session
OpenGL textures
drawn by Unity
original DASH client
MPD-driven segment fetching; same file and export names
removed from the build
Constraints handled outside the decoder
• fixed atlas size per session: set at encode time
• atlas change between sequences: decoder restarted between conditions
• no gaps inside a GOF: only complete GOFs are pushed
(b) Teardown fix in stopVideoDecoders()
upstream
host thread
finish()
stop(): join
blocks forever, host freezes
decoder thread
parked on an empty input, never sees the finish flag
patched: 4 lines, mirrors the audio path
host thread
finish()
push empty packet
stop(): join
returns
decoder thread
wakes, exits
stress test: 10 sessions (5 starved), 9 restarts, 6 min idle: no hangs
new (this work)
upstream, patched by us
upstream / third party, unmodified